Optical Remote Sensing Image Change Detection Based on Attention Mechanism and Image Difference

Xueli Peng, Ruofei Zhong, Zhen Li, and Qingyang Li

***Abstract*— This study presents a new end-to-end change detection network, called difference-enhancement dense-attention convolutional neural network (DDCNN), that is designed for detection of changes in the bitemporal optical remote sensing images. To model the internal correlation between high-level and low-level features, a dense attention method consisting of several up-sampling attention units is proposed. Both the up-sampling spatial and up-sampling channel attention are adopted by the unit. The unit, which can use high-level features with rich category information to guide the selection of low-level features, can use the spatial context information to capture the changed features of ground objects. Furthermore, DDCNN also pays attention to the differentiating features of the bitemporal images. By introducing a DE unit, each pixel is weighted and the features are selectively aggregated. The combination of dense attention and the DE unit improves the effectiveness of the network and its accuracy in extracting the change features. The effectiveness of the proposed approach is demonstrated via five challenge data sets. The experimental results show that DDCNN achieves new state-of-the-art change detection performance on these five challenging data sets. For the seasonal change detection data set in particular, compared with the best existing change detection model, the proposed method increases the F1 score and IoU by 2.96% and 5.17%, respectively; compared with the baseline method, our method improved 3.75% and 6.50% on the F1 score and IoU, respectively.**

***Index Terms*— Attention mechanism, change detection, fully convolutional network, image difference, optical remote sensing image.**

I. Introduction

CHANGE detection is a fundamental but challenging task in remote sensing, whose goal is to automatically detect changes in land cover or land use in the same geographic area based on bitemporal or multitemporal images. In general, change detection plays an important role in urban planning [1], forest monitoring [2], disaster assessment [3]–[5], and the dynamic detection of land use [6]. With the development of earth observation technology, increasingly more sensors have been designed for earth observation, such as various satellite sensors (e.g., the sentinel series, the Gaofen series), and aircraft-mounted sensors. The abundance of data provides a basic guarantee for the various applications of remote sensing. It is easier to obtain images of the same geographical area at different times, making it possible to dynamically detect the changes in the earth's surface in near real-time.

In recent decades, change detection technology has been an active research area [9] in the field of remote sensing and many change detection methods have been proposed and applied. According to the basic unit, change detection methods can be divided into *pixel-based* and *object-based* methods [6].

A. Pixel-Based Methods

The pixel-based change detection algorithms generate a change map by comparing the coregistered image pairs of different temporal scales pixel by pixel. These algorithms usually include three processes. First, image preprocessing is conducted, and it includes radiation correction [4], [5], geometric correction [4], [5], brightness normalization [10], and so on. This step is conducted to eliminate errors due to the solar altitude angle, atmospheric conditions, and so on. Then, appropriate detection methods are selected to generate the difference map. Finally, a threshold or clustering method is used to binarize the detection results.

An appropriate method for the generation of difference maps is essential to effectively carry out a change detection task. These methods can be divided into three general categories, namely, image arithmetic operations (such as image difference [10], image ratio, regression analysis [11]), image transformation (e.g., vegetation index [5], change vector analysis (CVA) [12], [13], principal component analysis (PCA) [14], [15], independent component analysis (ICA) [16]), image classification [17], [18], and other advanced algorithms, such as slow feature analysis (SFA) [18]–[20] and iteratively reweighted multivariate alteration detection (IR-MAD) [21]. In practical applications, the results of a single method often contain noises and errors, and the requirements of production accuracy cannot be met. Therefore, many scholars have combined different change detection methods to improve the performance in specific tasks [4], [5], [14].

B. Object-Based Methods

The object-based change detection methods utilize spectrum, texture, and other information to segment the raw image into different categories, and then obtain the change map via

Manuscript received July 19, 2020; revised September 19, 2020 and October 2, 2020; accepted October 18, 2020. Date of publication November 10, 2020; date of current version August 30, 2021. This work was supported by the National Natural Science Foundation of China under Grant 42071444. *(Corresponding author: Ruofei Zhong.)*

The authors are with the Key Laboratory of 3D Information Acquisition and Application, Ministry of Education, Beijing Advanced Innovation Center for Imaging Technology, Academy for Multidisciplinary Studies, Capital Normal University, Beijing 100048, China (e-mail: xuelipeng@cnu.edu.cn; zrf@cnu.edu.cn; sdlz123@126.com; 779477922@qq.com).

Digital Object Identifier 10.1109/TGRS.2020.3033009

1558-0644 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

comparison. To detect sudden changes and subtle changes, Lefebvre *et al.* [22] combined the contour analysis of an object with the texture from the geometric point of view to evolve the analysis. Huang *et al.* [23] fused luminance and texture differences between the different frames, which were found to be robust to noise and illumination changes. Robertson and King [24] compared the pixel-based and object-based methods in land use/land cover (LULC) research. By combining image segmentation, image differencing, and statistical testing, Desclée *et al.* [2] proposed a segmentation method that could increase the change detection performance but was not suitable for pixel-wise change detection.

The pixel-based change detection algorithms emphasize the change in the spectral value of a single pixel, but do not consider the relationship between the adjacent pixels and ignore the spatial environment. Although the object-based change detection methods comprehensively consider the spatial environment and spectral information of raw images, they are sensitive to registration errors and objects shadows, which may limit the detection accuracy. Furthermore, both the pixel- and object-based change detection methods not only require abundant manual interference but also are easily affected by sensors and light conditions.

Deep learning in computer vision has been in continuous development and has achieved very outstanding performance in recent years. Many scholars have introduced deep learning technology to remote sensing processing and have achieved excellent performance in image classification [25], [26], semantic segmentation [27]–[29], object detection [30], [31], change detection [17], [32]–[35], and other tasks. For the change detection tasks, deep-learning-based methods blur the boundary between the traditional pixel- and object-based methods, as they predict pixel-wise classification maps and highly semantically abstract spatial contexts from raw images [17]. Compared with the traditional pixel- and object-based change detection methods, the methods based on deep learning do not require image preprocessing, which may not only reduce manual intervention but also can avoid errors due to preprocessing. The post-processing workload can also be reduced.

There exist two main types of change detection networks. The first type is the single-branch networks that take the concatenated coregistered image pairs as the input of the model and then directly generate the binary change map. Based on the theory that changed and unchanged pixels correspond to quickly and slowly varying features in SFA, Du *et al.* [19] applied SFA in a convolutional network to suppress the spectral difference between slowly varying unchanged pixels and strengthen the changed pixels. By combining deep belief networks and feature change analysis, Zhang *et al.* [32] proposed an unsupervised change detection technique to suppress irrelevant variations and highlight changes. Mou *et al.* [35] introduced a convolutional neural network and a recurrent neural network into an end-to-end network, which could generate spectral–spatial feature representations and reveal temporal dependence in bitemporal images. The second type is the double-branch networks with shared weights [36], [37] (i.e., the Siamese network). The Siamese network processes the coregistered image pairs simultaneously and maps them into a high-dimensional space. A variation intensity map is then obtained by measuring the pixel distances of the high-dimensional feature maps of multitemporal images. Similar to the pixel-based approach, the variation intensity maps require threshold segmentation and cluster analysis.

Most of the algorithms can detect the change features well, but they lack a focus on the extraction of the overall features and attention to the target change areas. The difference maps of the original bitemporal image pairs have not been fully used either.

Remote sensing images have the peculiarity of complex scenes and abundant objects. Even coregistered image pairs have various deviations, such as position deviations and spectral differences. Position deviations, which are caused by registration errors or geometric correction errors, prevent the same feature from completely coinciding. Spectral differences, which are caused by factors such as the solar altitude angle, illumination conditions, and seasonal changes, result in the same objects having quite different spectra. Traditional algorithms cannot meet the needs of large-scale change detection, and the existing deep-learning-based networks rarely pay attention to the spatial context information and the internal connections between high-level and low-level features. In this article, we propose a novel change detection network, called difference-enhancement dense-attention convolutional neural network (DDCNN). The dense attention can assist the convolutional neural network (CNN) to improve the ability of texture and detail extraction. Moreover, the DE unit can model the temporal difference, which can extract the change density map and filter change noise. In summary, the main contributions can be summarized as follows.

1) This article presents a powerful and general end-to-end network architecture for the change detection in both homogeneous and heterogeneous optical images.
2) A dense attention method that pays close attention to each stage of feature fusion is proposed. In each up-sampling attention (UA) unit, up-sampling spatial and channel attention (CA) are adopted to simultaneously capture change information in both spatial and channel dimensions. The UA unit can better locate detailed information and texture features.
3) A DE unit is proposed. In combination with the difference image, the DE unit is designed to process the bitemporal intuitive change maps and can map the difference image into a new feature space, thereby fully excavating the change information and generating a change density map.

The rest of this article is organized as follows. Section II reviews the related work. Section III describes the proposed method. In Section IV, experiments are designed to evaluate the proposed method. Finally, Section V presents the conclusion of this research.

II. Related work

In this section, the related research is reviewed.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

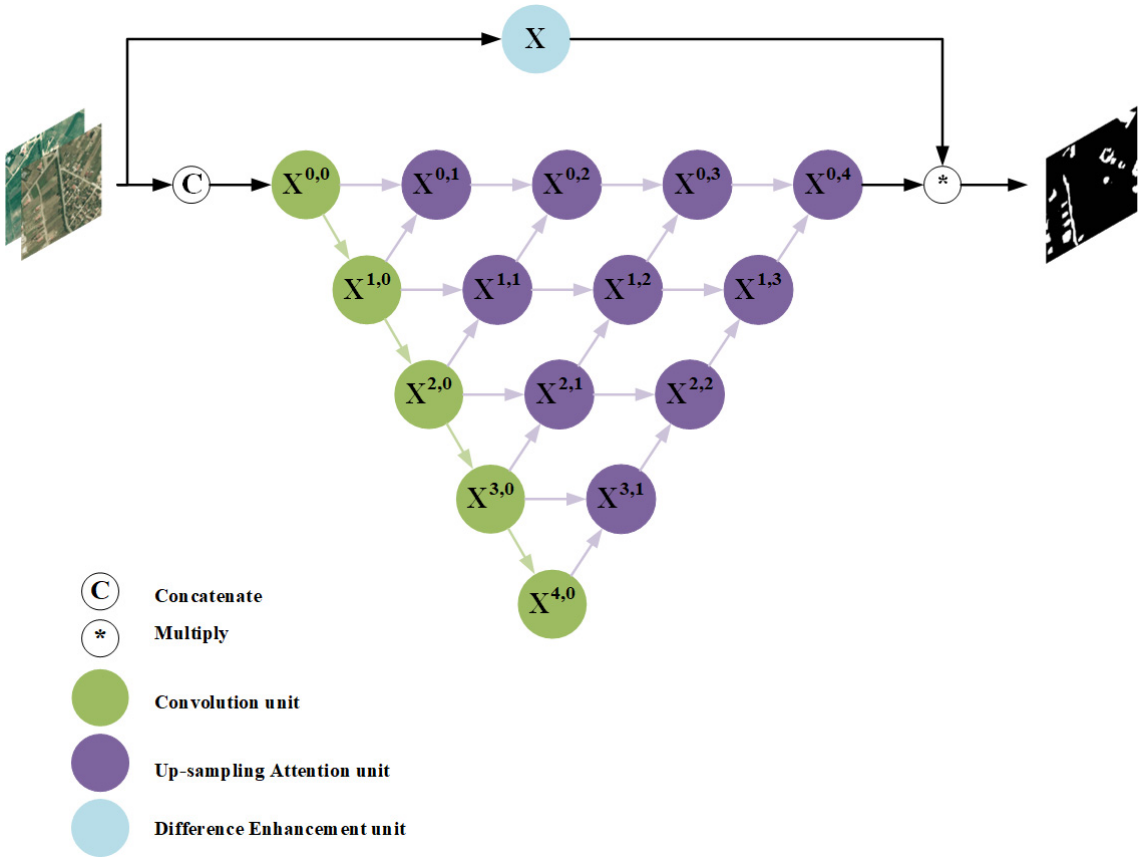


Fig. 1. DDCNN.

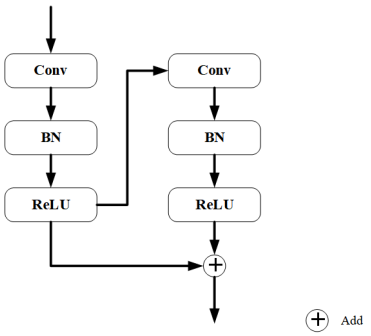


Fig. 2. Convolution unit.

A. Change Detection Networks With Attention Mechanism

Recently, the attention mechanism has been widely used in natural language processing [38], speech recognition, computer vision [39], [40], and other fields. The attention mechanism is an observation mechanism that imitates the human eye and selectively pays attention to a portion of all information while ignoring other visible information. Via an attention mechanism, a network can automatically weight the feature maps to enhance the change characteristics, which greatly promotes the efficiency and accuracy of the network.

To date, some scholars have applied the attention mechanism to change detection [41], [42]. These methods use an attention mechanism to enhance the representation of change features extracted by the backbone. Chen *et al.* [41] increased the robustness of the pseudo-change information by introducing both spatial and channel attention into the convolution network [7]. Liu *et al.* [42] introduced the attention mechanism after an information transmission module to enhance the change information. These methods with attention mechanisms use the attention mechanism to enhance the change characteristics.

B. Change Detection Networks Based on U-Net Series

U-Net series have excellent performance in segmentation tasks (such as supervised medical segmentation) due to their unique skip connection and lightweight network architecture. Similar to the medical image data sets, the existing change detection data sets for remote sensing are often small. Therefore, many scholars have made improvements based on the U-Net series and proposed the networks for change detection.

By combining the multirate Atrous convolutions to expend the receptive field, Zhang *et al.* [17] proposed an object-based change detection network to classify landcover and then generated change maps by comparing the predicted classification map with the geographic information system (GIS) map. Papadomanolaki *et al.* [43] proposed a deep framework for temporal modeling that combines a fully convolutional network and the long-short-term-memory (LSTM). To fuse the change maps from different levels, Peng *et al.* [44] proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

a change detection network with a fusion strategy of multiple side outputs based on nested U-Net. Pyramid pooling with different convolutional kernels can learn multiscale features; in their research, Lei *et al.* [3] introduced pyramid pooling to a fully convolutional network for landslide mapping to overcome the complex characteristics and spatial uncertainties of the landslides. In the work by Daudt *et al.* [8], a fully convolutional network used as a single network is also used as the Siamese networks, namely, FC-Siam-conc and FC-Siam-diff. In the down-sampling stage, both FC-Siam-conc and FC-Siam-diff use a Siamese network with shared weights; in the up-sampling stage, FC-Siam-conc fuses the features via concatenation, while FC-Siam-diff fuses feature by feature difference.

There also exist some change detection methods based on U-Net used in the field of computer vision. For example, motivated by the need for more frequent and efficient updates to the large-scale maps used in autonomous vehicle navigation, Alcantarilla *et al.* [45] performed structural change detection in street-view videos captured by a vehicle-mounted monocular camera over time.

III. Methodology

In this section, a general framework of the proposed network is first presented. The UA unit that makes up the dense attention and simultaneously captures long-range contextual information in the spatial and channel dimensions is then presented. Finally, the DE unit is described.

A. Network Architecture

Change detection in remote sensing can be understood as the segmentation of changes from optical images of the same geographical area taken at different times. The improved change detection models based on segmentation networks such as U-Net [46] and UNet++ [47] have excellent consistency in terms of the size of the extracted features and the input images. These networks have strong feature extraction and detail retention capabilities. Among them, UNet++_MSOF [44] can accurately detect the change information and can also better retain the edges to avoid the aliasing effect. Furthermore, due to the inherent advantage of UNet++ in feature extraction, UNet++_MSOF can integrate high-level semantic information and low-level texture features, thereby improving the accuracy of feature extraction. Inspired by this advantage, an improvement of UNet++_MSOF is made in this work. The architecture of the proposed DDCNN, as shown in Fig. 1, is composed of three components, namely, the convolution unit, the UA unit, and the DE unit. The convolution unit is a simplified residual block and is shown in Fig. 2.

To improve the effect of change detection, an attention mechanism is applied in the proposed network to extract more rich details and texture information. In previous work, researchers used the attention mechanism as a portable module for the final enhancement of the features extracted by the backbone [41], [42]. It is believed that this is not sufficient to enhance the features extracted by the backbone only. Therefore, the method of dense attention is proposed, which fuses and extracts the change features in each stage via the UA unit. The UA unit is used in the up-sampling/deconvolution stage. The dense attention method can fully explore the intrinsic correlation between high-level and low-level features. The use of dense attention better preserves the texture and detail information of the changing area in the bitemporal remote sensing images, which not only increases the antiinterference ability but also improves the ability to detect the changes in small objects.

Unlike ordinary images, remote sensing images are the products of earth observations from satellites or space shuttles. There are various objects in the images, such as buildings, roads, forests, farmlands, and cities, which make the scenes more complicated in the remote sensing images. In neural networks, an appropriate increase in the number of convolution kernels can improve the feature extraction ability of the network and enhance the fitting performance of the model. To better model the complex scenes of remote sensing images, the number of convolution kernels in each layer is increased to {64, 128, 256, 512, 1024}, which increases the performance of the network. However, after doubling the number of convolution kernels, although the feature extraction ability of the network is improved, the amount of network computation is greatly increased, thereby slowing down the model calculation. Therefore, the number of short connections is further reduced, which not only greatly reduced the calculation but also did not show a significant drop in feature extraction.

The image difference is a traditional pixel-based change detection method. Because the change map generated by image differences often contains noise and pseudo changes, an image DE unit is proposed by combining the image difference and the convolutional layers.

As shown in Fig. 1, the processing flow of DDCNN is mainly divided into two components. The first component concatenates the bitemporal image pairs as the input of the backbone network and extracts the accurate change features from the original images via the convolution unit and the UA unit. The second component inputs the difference images of the bitemporal image pairs into the DE unit to generate the change intensity maps. Formally, each unit is calculated as follows: let $\mathbf{Y^{i,j}}$ be the output of each node $\mathrm{X}^{i,j}$ where i represents the downsampling layer and j represents the jth UA node of layer i ($j = 0$ indicates that the node is a convolutional unit). Let $\mathbf{Y^{res}}$ be the output of the DE unit. The formulas are described as follows:

$$Y^{i,j} = \begin{cases} C([Y^{t1}, Y^{t2}]), & i = 0, j = 0 \\ C(Y^{i-1,j}), & i > 0, j = 0 \\ \mathrm{Att}(Y^{i,j-1}, Y^{i+1,j-1}), & j != 0 \end{cases} \quad (1)$$

$$Y^{\mathrm{res}} = \mathrm{res}(Y^{t1} - Y^{t2}) \quad (2)$$

where the function $C(\cdot)$ is the convolutional unit operation, $\mathrm{Att}(\cdot)$ is the UA unit operation, $\mathrm{res}(\cdot)$ is the simplified residual block operation in the DE unit, $\mathbf{Y^{t1}}$ and $\mathbf{Y^{t2}}$ indicate the images before and after the change, respectively, and [] represents the concatenation operation. Specifically, the node at $\mathrm{X}^{0,0}$ receives the bitemporal image pairs as the input; the node at $\mathrm{X}^{i,j}$ ($i > 0$, $j = 0$) takes the output of node $\mathrm{X}^{i-1,j}$ as the input; the note at

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

$X^{i,j}$ ($j > 0$) takes the results of node $X^{i,j-1}$ and note $X^{i+1,j-1}$ as the input.

In the proposed network, the outputs of the nodes in the same layer have the same size. In the down-sampling stage, when $i > 1$, the size of the output feature of each convolution unit compared with the input feature map is halved, and the number of channels is doubled. In the up-sampling stage, there are two inputs in the UA unit; however, the size of the output feature map depends only on that of the low-level features of the input. Taking node $X^{0,1}$ as an example, the two inputs are $\mathbf{Y^{0,0}}$ ($32 \times 256 \times 256$) and $\mathbf{Y^{1,0}}$ ($64 \times 128 \times 128$), so the output of the node is $\mathbf{Y^{1,1}}$ with dimensions of $32 \times 256 \times 256$.

B. UA Unit

A method of dense attention is proposed for the decoding stage and is realized by the UA unit (Fig. 3), which can model the rich context of the local features in each stage. The high-level features with rich category information are used as guidance for the selection of low-level features for category localization. The high-level features are also used to weight the low-level features and to provide precise pixel-level concentration for the extraction of detailed features. There are two forms of attention mechanism. One is the CA mechanism. CA performs global average pooling on the features in each channel but ignores the local information in each channel to extract multiscale features effectively. The other is the spatial attention (SA) mechanism, which ignores the channel domain information and treats the features in each channel equally. The proposed attention unit was designed by combining the two attention mechanisms.

The unit is divided into two parts, namely, up-sampling spatial attention (USA) and up-sampling channel attention (UCA). In detail, when calculating the SA map, we first perform a 3×3 convolution on the low-level and high-level features, respectively. Then we multiply the two features pixel by pixel. Finally, after a 3×3 convolution and a sigmoid activation function, the SA map is generalized. The CA map is obtained by 3×3 convolution after the global average pooling. After preparing the attention maps, the low-level features undergo a simple 3×3 convolution, and the attention maps are multiplied pixel by pixel to achieve the ability to simultaneously capture the spatial context from the channel and spatial domains.

It should be noted that before computing the SA map, the high-level features are up-sampled to ensure that the size of the high-level features is the same as that of the low-level features. However, the up-sampling of the CA map is not required.

C. DE Unit

Although the difference map is the most intuitive means to reveal the changes in bitemporal images, the existence of spectral and position errors prevents the difference map from being directly used to represent the change. Therefore, the DE unit (Fig. 4), which is a flexible structure that can be removed at any time, is introduced. The DE unit is used to connect DDCNN from the beginning to the end. In detail, the difference map of the bitemporal image pairs is first obtained, and it is convolved. In the convolution process, a short connection, like ResNet, is adopted. Then, the obtained abstract features of the difference maps are used to weight the features extracted by the raw bitemporal images.

IV. Experiments

The proposed approach was evaluated on five change detection data sets: the season-varying data set [48], the WHU building change detection data set [1], the SZTAKI AirChange benchmark set [49], the Onera Satellite change detection (OSCD) data set [50], and the LEVIR-CD data set [51]. The experimental results demonstrate that DDCNN achieved state-of-the-art performance on all data sets. In this section the data sets and implementation details are first introduced. Then the results of a series of ablation experiments on the season-varying change detection data set are reported. The results on the five data sets are then, respectively, presented.

A. Data Sets and Implementation Details

1) Data Sets: Season-Varying Change Detection Data Set: The season-varying change detection data set was obtained by Google Earth (Digital Globe). The data set consists of seven image pairs of 4725×2200 pixels and four image pairs of 1900×1000 pixels. The spatial resolution ranges from 3 to 100 cm per pixel, and the seasons vary widely among the bitemporal images. As this data set contains changes in different objects (such as buildings, roads, vehicles) and ignores the changes caused by seasonal differences, brightness, and other factors, it poses a great challenge to change detection algorithm. The data sets, generated by cropping and rotating, contain 10 000 training sets, 3000 test sets, and 3000 validation sets.

WHU Building Change Detection Data Set: The WHU building change detection data set consists of images from satellites, including QuickBird, Worldview series, IKONOS, and ZY-3. The images in the data set cover the area of the earthquake that struck Christchurch, New Zealand, in February 2011. The data set contains a pair of coregistered images taken in 2012 and 2016, with change detection labels for the buildings. The size of each image is 32507×15354 pixels, and the spatial resolution is 0.2 m per pixel. The data set was divided into 256×256 pixel patches without overlapping, and invalid patches were filtered out (no change and all change). The data set was then divided according to the ratio of 7:2:1 to obtain 3167 training sets, 935 validation sets, and 436 test sets.

OSCD Data Set: The OSCD data set comprises 24 pairs of multispectral images taken by the Sentinel-2 satellites between 2015 and 2018. The images vary in spatial resolution between 10, 20, and 60 m. There are 13 bands in the images captured by Sentinel-2, and the red, blue, and green bands were used in this work. By overlapping the adjacent image patches by 50%, the data set was clipped into patches with the sizes of 256×256 pixels, and each patch was then flipped. Finally, the patches before and after flipping were rotated by 90°, 280°, and 270°. The patches were divided according to the ratio of 7:2:1, and finally 1288 training sets, 368 validation sets, and 184 test sets were obtained.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

SZTAKI AirChange Benchmark Set: The SZTAKI AirChange Benchmark (SZTAKI) set contains 13 pairs of aerial images with sizes of 952 × 640 pixels, and the resolution is 1.5 m/pixel. The SZTAKI data set takes the new built-up regions, buildings, large groups of trees, and fresh plow-land into consideration. The SZTAKI data set was processed in the same manner as the OSCD data set, and 1271 training sets, 363 validation sets, and 182 test sets were ultimately obtained.

LEVIR-CD data set: The LEVIR-CD data set consists of 637 image pairs with a size of 1024 × 1024 pixels, which were collected from Google Earth. This data set used for building change detection was divided into training, validation, and test sets by the author. On this basis, we cut each image pair into 256 × 256 pixel patches without overlapping. After filtering out the invalid patches, we ultimately obtained 3169 training sets, 438 validation sets, and 937 test sets.

2) Experimental Settings: In the experiments, the cross-entropy was applied as the loss function. The Adam with learning rate 1e-4, momentum 0.9, and weight decay 5e-4 was chosen as the optimization algorithm. It is necessary to point out that the batch size of the training data was set to 4, except for the part of the experiment using DSANet [41] (batch size 6 for the season-varying change detection data set and SZTAKI data set). The same training and testing settings were adopted on all data sets. Moreover, the proposed network was implemented based on the Pytorch framework, and a single NVIDIA Quadro TITAN XP GPU with 12 G memory is used for training and testing.

3) Comparative Method and Evaluation Metrics: To verify the validity of the proposed network, it was compared with the commonly used change detection model based on the U-Net series and some recently developed change detection models. FC-EF [8] takes the bitemporal image pairs concatenated as the input. FC-Siam-diff [8] and FC-Siam-conc [8] are the extensions of FC-EF; in the encoding stage, these two networks use a Siamese network with shared weights. BiDataNet [43] adds LSTM to the U-Net. CDNet [45] was designed for street view change detection. The UNet++_MSOF [44] is used as the baseline of the proposed method. DASNet [41] applies the attention mechanisms to a Siamese network.

Precision (P), recall (R), F1 score, intersection over union (IoU), and overall accuracy (OA) are used to quantitatively evaluate the performance of the proposed method. These metrics are calculated as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{3}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{4}$$

$$F1 = \frac{2PR}{P+R} \tag{5}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{6}$$

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{7}$$

where true positive (TP) is the number of nonchange pixels detected correctly, false positive (FP) is the number of nonchange pixels unpredicted, true negative (TN) is the number of change pixels detected correctly, and false negative (FN) is the number of change pixels unpredicted. Among the indexes above, the F1 score and IoU could better reflect the generalization ability of a model.

Precision represents the probability of true change in all pixels detected as having changed. Recall indicates the probability that all changed pixels are correctly detected. It is difficult to simultaneously achieve high recall and high precision. However, the F1 score is the harmonic mean of recall and precision, which can balance the conflict by considering both precision and recall simultaneously. The IoU is the area of overlap between the predicted changed pixels and the changed pixels divided by the union area between them. In summation, the F1 score and IoU can better reflect the change detection ability of a model.

TABLE I
ABLATION EXPERIMENT ON THE SEASON-VARYING CHANGE DETECTION DATA SET

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
UNet++_MSOF	94.73	87.02	90.71	83.01	97.76
CNN-USA	96.16	91.23	93.63	88.02	98.44
CNN-UCA	95.33	92.26	93.77	88.27	98.46
CNN-UA	95.37	**92.80**	94.07	88.80	98.53
CNN-UA-DE (ours)	**96.71**	92.32	**94.46**	**89.51**	**98.64**

B. Ablation Study

During up-sampling, the UA unit is used in the proposed method to capture long-range dependencies for better understanding of a scene. The DE unit is used to add additional weights to changed features. To evaluate the UA and DE units, a couple of ablation experiments were conducted on the season-varying change detection data set. In detail, we conduct experiments with several settings, including the USA, UCA, UA, and DE units.

As shown in Table I, the proposed method exhibited a remarkably improved performance of change detection. Compared with the baseline (UNet++_MSOF), our method, which brings a score of 3.75% and an IoU of 6.50% improvement, yields a result of 94.46% in the F1 score and 89.51% in IoU. Specifically, the UA achieves an F1 score of 94.07% and IoU 88.80%. It outperformed the IoU of USA and UCA by 0.78% and 0.53%, respectively, which confirms the effectiveness of extracting the attention map from the spatial and channel domains at the same time. Moreover, the DE unit improved the model by 0.39% and 0.71% in terms of the F1 score and IoU, respectively.

C. Season-Varying Change Detection Data Set

We carry out experiments on the season-varying change detection data set to evaluate the effectiveness of DDCNN. The quantitative results of the validation set are presented in Table II. DDCNN was compared with other latest methods and it exhibited significant improvement of 2.96% and 5.17% in terms of the F1 score and IoU, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

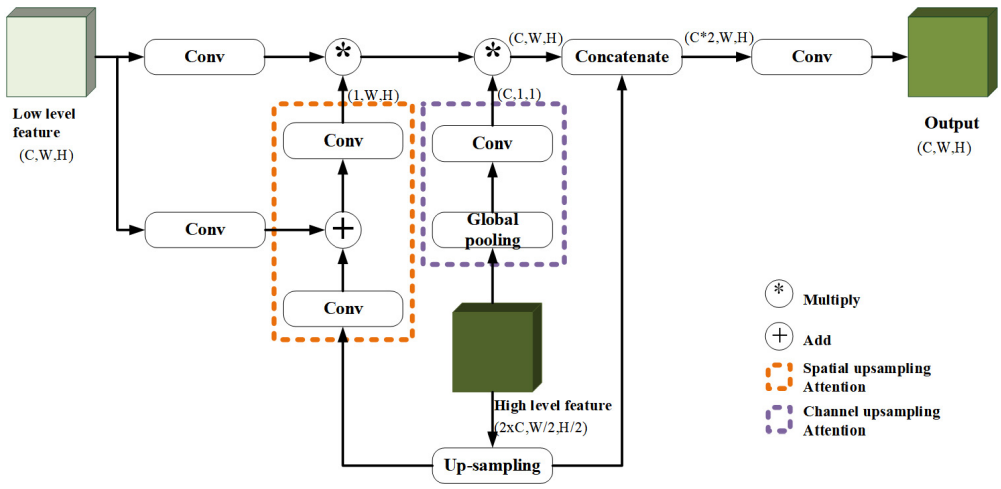


Fig. 3. UA unit.

TABLE II
RESULTS ON THE SEASON-VARYING CHANGE DETECTION DATA SET

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
FC-EF	70.62	64.42	67.38	50.80	92.16
FC-Siam-diff	91.95	55.37	69.11	52.81	93.79
FC-Siam-conc	92.91	52.70	67.25	50.66	93.55
BiDataNet	95.94	87.46	91.50	84.34	97.96
CDNet	92.10	78.19	84.57	73.27	96.42
UNet++_MSOF	94.73	87.02	90.71	83.01	97.76
DASNet	91.35	87.64	89.46	80.93	97.41
ours	**96.71**	**92.32**	**94.46**	**89.51**	**98.64**

TABLE III
RESULTS ON THE WHU BUILDING CHANGE DETECTION DATA SET

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
FC-EF	78.86	78.64	78.75	64.94	93.03
FC-Siam-diff	84.73	87.31	86.00	75.44	95.33
FC-Siam-conc	78.73	88.80	83.47	71.62	94.22
BiDataNet	86.75	**90.60**	88.63	79.59	96.19
CDNet	91.75	86.89	89.26	80.60	96.57
Unet++_MSOF	91.96	89.40	90.66	82.92	96.98
DASNet	88.23	84.62	86.39	76.04	95.30
ours	**93.71**	89.12	**91.36**	**84.09**	**97.23**

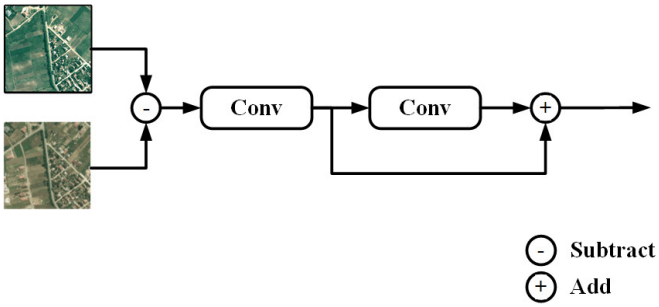


Fig. 4. DE unit.

The season-varying change detection data set contains several scenes, and three typical scenes were chosen for visualization. Figs. 5–7 show, respectively, the change detection results of buildings, vehicles, and road scenes. For building changes, most algorithms were able to extract the changed buildings, but DDCNN achieved the highest detection accuracy regardless of whether it is covered with snow. Fig. 5 reveals that DDCNN made the fewest false detections and missed detections of the changed buildings. Regarding vehicle change detection (see Fig. 6), our network could detect almost all the change vehicles. Although the boundaries detected by DDCNN are slightly larger than the real labeled boundaries, the vehicles were still well-distinguished from each other. Compared with other algorithms, the changed vehicles detected by DDCNN were more accurate. As shown in Fig. 7, DDCNN could detect the changing roads, especially the narrow roads, even if they were covered by snow (row 4 in Fig. 7).

Although the results of DDCNN were more accurate than those of other methods, there were still some pixels that were falsely detected or undetected. This is mainly due to the substantial variations in the scenes in this data set and the heavy disturbance of snow. But compared with the other methods, DDCNN could better overcome the effect of snow. According to Figs. 5–7, DDCNN performed better in terms of detail extraction and patch integrity; thus, it successfully overcame the interference of snow and has strong robustness.

D. WHU Building Change Detection Data Set

We also conduct experiments on the WHU building change detection data set. The comparison between the performance of the proposed method and the existing state-of-the-art methods are shown in Table III. The baseline (UNet++_MSOF) yielded F1 score 90.66% and IoU 82.92%, while DDCNN boosts the performance to F1 score 91.36% and IoU 84.09%.

It can be seen from the first two rows in Fig. 8 that all the algorithms can detect the changed buildings. However, while the other algorithms exhibited false or missed detection,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

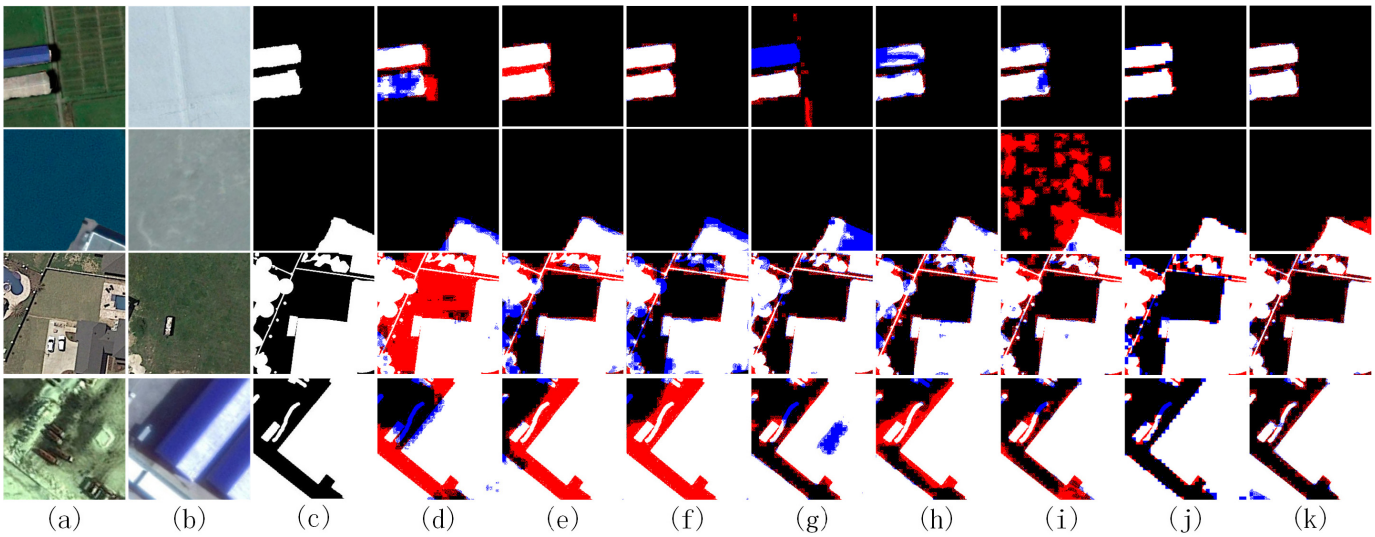


Fig. 5. Results of buildings on the season-varying change detection data set. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) BiDataNet. (h) CDNet. (i) UNet++_MSOF. (j) DASNet. (k) Ours. Black indicates the no change detected correctly. White indicates the changes detected correctly. Red indicates the changes detected by error. Blue indicates unpredicted changes.

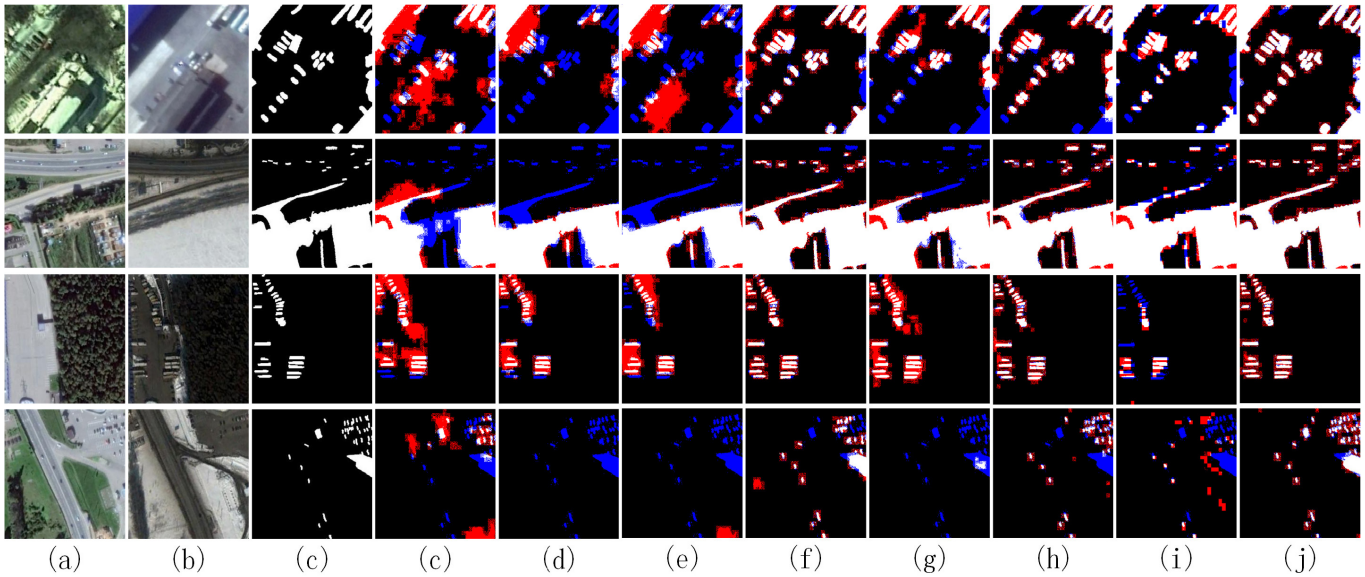


Fig. 6. Results of vehicles on the season-varying change detection data set. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) BiDataNet. (h) CDNet. (i) UNet++_MSOF. (j) DASNet. (k) Ours. Black indicates the no change detected correctly. White indicates the changes detected correctly. Red indicates the changes detected by error. Blue indicates unpredicted changes.

the results of DDCNN were more accurate and less noisy. Besides, the proposed method sufficiently overcame the effects of noise (nonbuilding changes such as road changes) in this data set, as shown in rows 3 and 4 in Fig. 8.

The experimental results show that DDCNN achieved a good performance on the WHU building change detection data set and exhibited a strong antiinterference ability.

E. OSCD Data Set

Experiments were conducted on the OSCD data set to verify the generalization of our proposed network. The OSCD data set is the only data set used in this work created entirely by satellite imagery. The quantitative results are shown in Table IV, and the visualization is shown in Fig. 9. The comparison results show that DDCNN has reached the optimal level on five indexes. Specifically, the precision, recall, F1 score, IoU, and OA values were 96.11%, 89.73%, 92.81%, 86.59%, and 99.66%, respectively.

TABLE IV
RESULTS ON THE OSCD DATA SET

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
FC-EF	85.27	75.11	79.87	66.49	99.06
FC-Siam-diff	92.67	55.59	69.49	53.25	98.79
FC-Siam-conc	88.76	66.40	75.97	61.25	98.96
BiDataNet	93.44	87.66	90.46	82.58	99.54
CDNet	87.64	81.43	84.42	73.04	99.26
UNet++_MSOF	93.03	86.29	89.53	81.05	99.50
DASNet	67.08	54.11	59.90	42.75	98.20
ours	**96.11**	**89.73**	**92.81**	**86.59**	**99.66**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

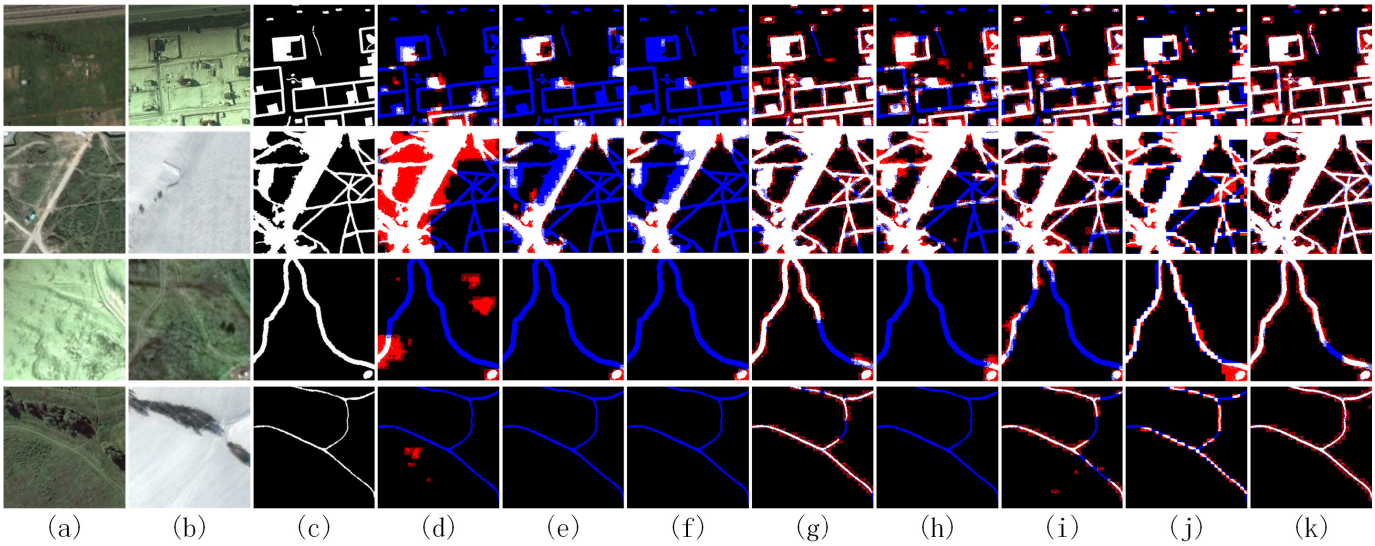


Fig. 7. Results of roads on the season-varying change detection data set. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) BiDataNet. (h) CDNet. (i) UNet++_MSOF. (j) DASNet. (k) Ours. Black indicates the no change detected correctly. White indicates the changes detected correctly. Red indicates the changes detected by error. Blue indicates unpredicted changes.

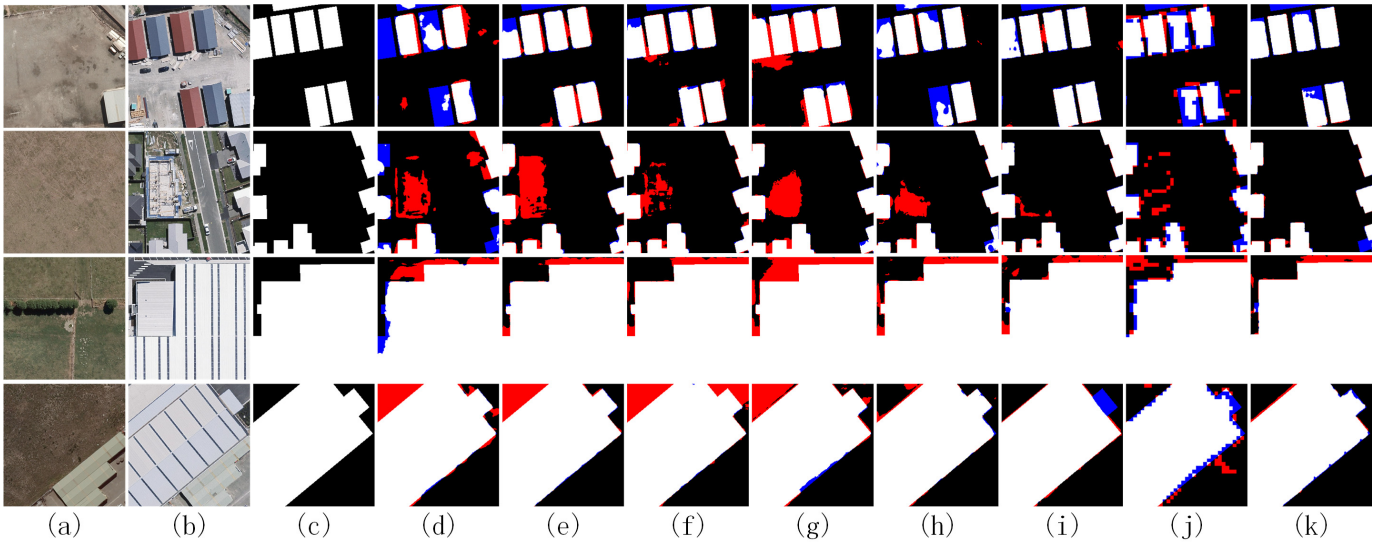


Fig. 8. Results on the WHU building change detection data set. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) BiDataNet. (h) CDNet. (i) UNet++_MSOF. (j) DASNet. (k) Ours. Black indicates the no change detected correctly. White indicates the changes detected correctly. Red indicates the changes detected by error. Blue indicates unpredicted changes.

There were discontinuous situations in the detection of changed narrow roads (row 4 in Fig. 9), which were mainly due to the lower resolution of Sentinel-2 and the complexity of the urban area. But for most areas, DDCNN can sufficiently detect the changed areas, as presented in Fig. 9. It is therefore evident that DDCNN can achieve excellent performance in the change detection and detail retention of small objects.

F. SZTAKI AirChange Benchmark Set

Experiments were carried out on the SZTAKI data set to further evaluate the effectiveness of the proposed method. As reported in Table V, DDCNN achieved 94.94% and 90.37% in F1 score and IoU, respectively, and outperformed the comparison methods by a large margin. Several results of different scenes, such as buildings, new built-up regions, fresh plowland, and a large group of trees, were selected for visualization, as shown in Fig. 10. The figure reveals that DDCNN better detected the changes in aerial images.

TABLE V
RESULTS ON THE SZTAKI AIRCHANGE BENCHMARK SET

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
FC-EF	89.86	87.11	88.46	79.31	98.65
FC-Siam-diff	91.38	71.89	80.47	67.32	97.92
FC-Siam-conc	90.67	72.05	80.30	67.08	97.89
BiDataNet	**94.88**	84.68	89.49	80.99	98.82
CDNet	92.53	88.84	90.65	82.89	98.91
UNet++_MSOF	94.74	90.72	92.68	86.37	99.15
DASNet	87.00	85.39	86.19	75.73	98.22
ours	94.79	**95.09**	**94.94**	**90.37**	**99.40**

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

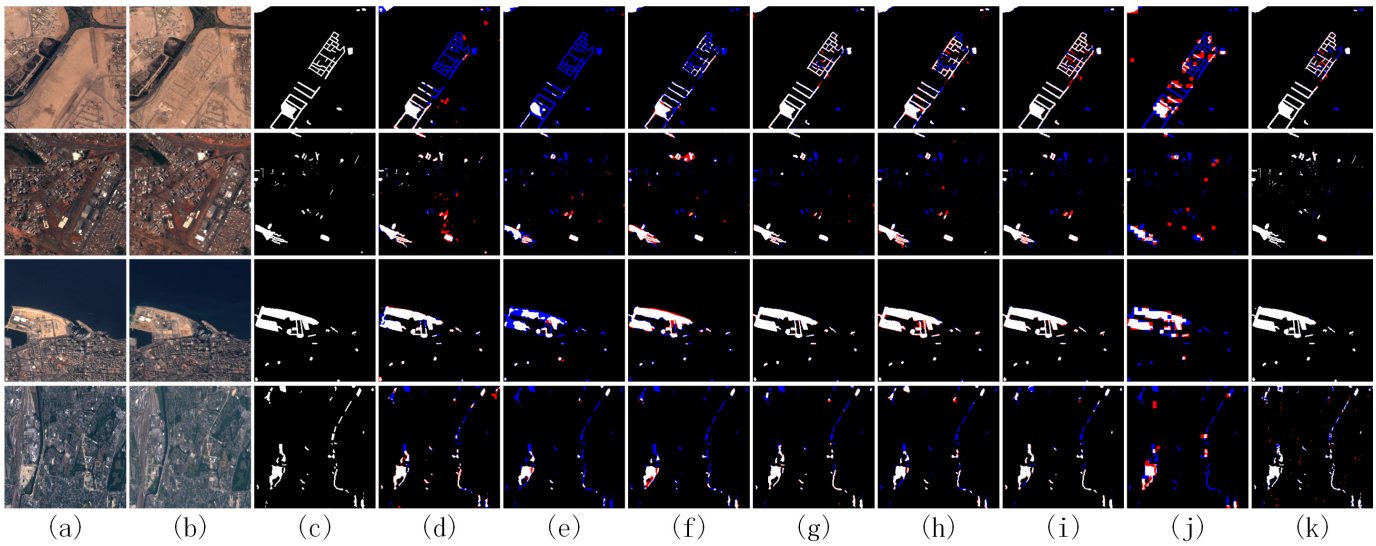


Fig. 9. Results on the OSCD data set. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) BiDataNet. (h) CDNet. (i) UNet++_MSOF. (j) DASNet. (k) Ours. Black indicates the no change detected correctly. White indicates the changes detected correctly. Red indicates the changes detected by error. Blue indicates unpredicted changes.

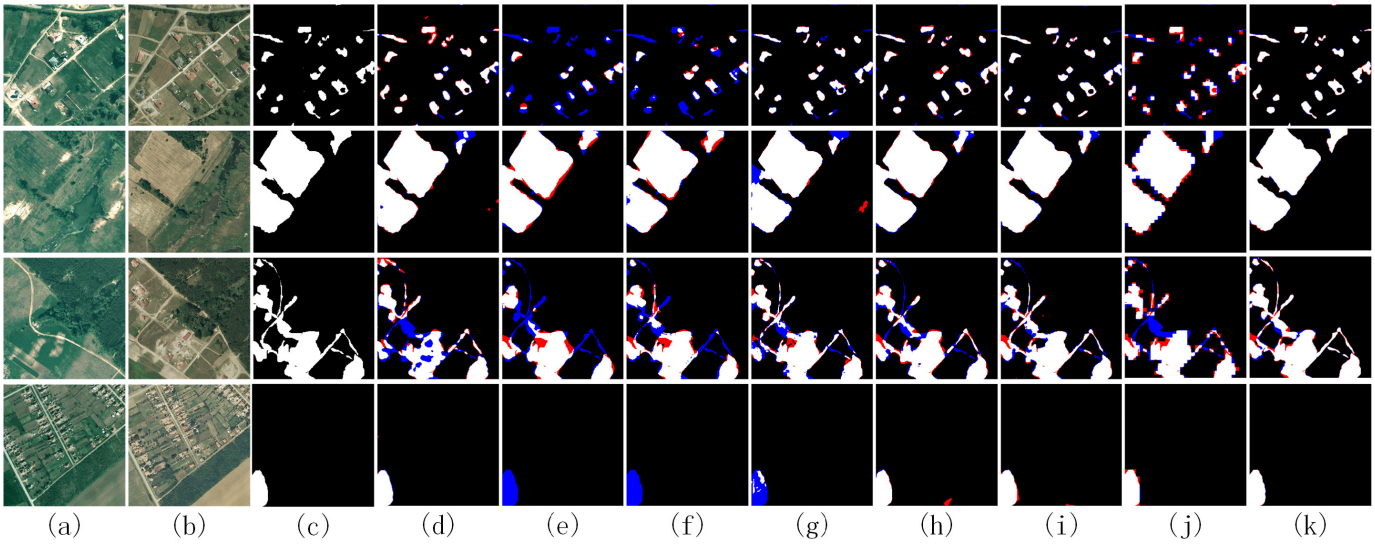


Fig. 10. Results on the SZTAKI AirChange benchmark set. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) BiDataNet. (h) CDNet. (i) UNet++_MSOF. (j) DASNet. (k) Ours. Black indicates the no change detected correctly. White indicates the changes detected correctly. Red indicates the changes detected by error. Blue indicates unpredicted changes.

TABLE VI
RESULTS ON THE LEVIR-CD DATA SET

Method	P (%)	R (%)	F1 (%)	IoU (%)	OA (%)
FC-EF	79.20	81.72	80.44	67.27	96.08
FC-Siam-diff	89.92	**89.00**	89.45	80.92	97.93
FC-Siam-conc	91.46	88.05	89.72	81.36	98.01
BiDataNet	**92.50**	87.14	89.74	81.39	98.04
CDNet	90.73	86.82	88.73	79.75	97.83
UNet++_MSOF	92.18	87.84	89.96	81.75	98.07
DASNet	81.10	78.81	79.94	66.59	95.59
ours	91.85	88.69	**90.24**	**82.21**	**98.11**

G. LEVIR-CD Data Set

Finally, we carried out experiments on the LEVIR-CD data set. The quantitative results are presented in Table VI. The proposed method, which, respectively, achieves 90.24% and 82.21% in the F1 score and IoU, and it outperforms the comparison methods by a slight margin. The visualization is shown in Fig. 11.

V. CONCLUSION

In this article, we propose a network architecture (DDCNN) for bitemporal image change detection. The network combines the method of dense attention and a DE unit to automatically learn the spectral information and adaptively extracts change features with spatial context. Specifically, in the UA unit, USA and UCA are combined to enhance the extraction of texture and detail information. The UA unit uses high-level features to simultaneously capture global dependencies on low-level features in the channel and spatial dimensions. In the DE unit, convolution is directly carried on the difference images to extract the high-level difference features. The experimental results demonstrate that the DDCNN is very robust. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

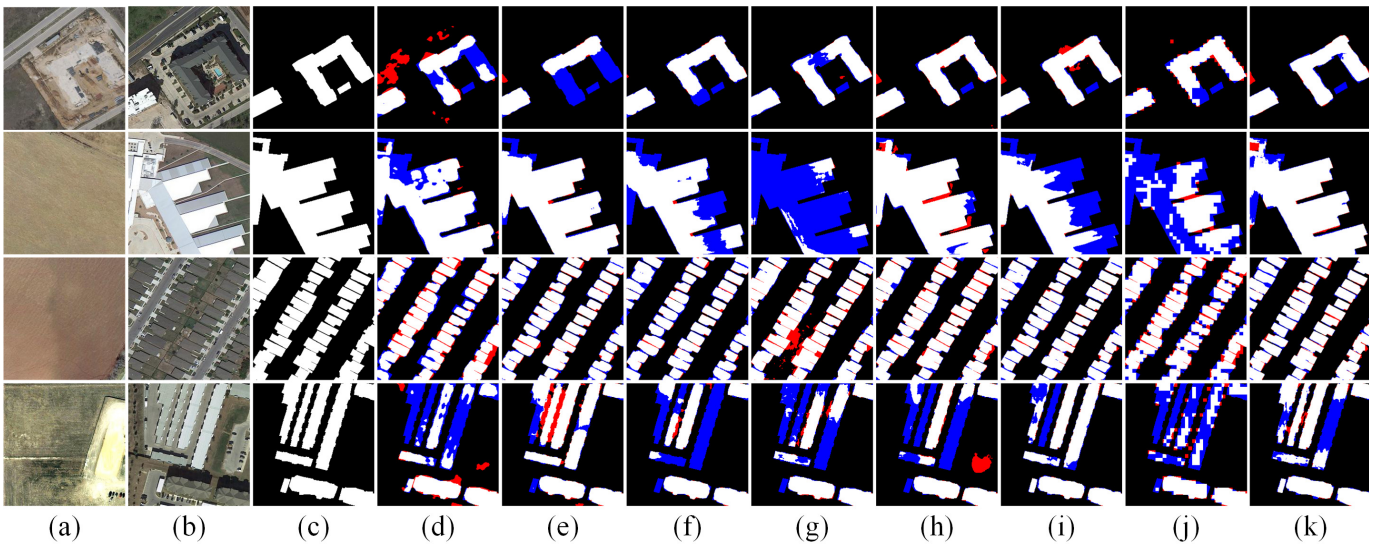


Fig. 11. Results on the LEVIR-CD data set. (a) Image T1. (b) Image T2. (c) Ground truth. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) BiDataNet. (h) CDNet. (i) UNet++_MSOF. (j) DASNet. (k) Ours. Black indicates the no change detected correctly. White indicates the changes detected correctly. Red indicates the changes detected by error. Blue indicates unpredicted changes.

combination of the dense attention and DE unit can effectively capture the spatial context of the bitemporal images. The results of change detection are also more accurate, especially in the extraction of textures and details. The results show that our network is valid for homogeneous, heterogeneous, and mixed source data. The DDCNN has achieved excellent consistency on the seasonal change detection data set, the WHU building change detection data set, the SZTAKI AirChange benchmark set, and the OSCD data set.

ACKNOWLEGEMENT

The authors would like to appreciate the authors of opening data sets for their contributions.

REFERENCES

[1] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[2] B. Desclée, P. Bogaert, and P. Defourny, "Forest change detection by statistical object-based method," *Remote Sens. Environ.*, vol. 102, nos. 1–2, pp. 1–11, May 2006.

[3] T. Lei, Y. Zhang, Z. Lv, S. Li, S. Liu, and A. K. Nandi, "Landslide inventory mapping from bitemporal images using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 982–986, Jun. 2019.

[4] Z. Li, W. Shi, P. Lu, L. Yan, Q. Wang, and Z. Miao, "Landslide mapping from aerial photographs using change detection-based Markov random field," *Remote Sens. Environ.*, vol. 187, pp. 76–90, Dec. 2016.

[5] P. Lu, Y. Qin, Z. Li, A. C. Mondini, and N. Casagli, "Landslide mapping from multi-sensor data through improved change detection-based Markov random field," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111235.

[6] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013.

[7] E. Guo *et al.*, "Learning to measure change: Fully convolutional siamese metric networks for scene change detection," 2018, *arXiv:1810.09111*. [Online]. Available: http://arxiv.org/abs/1810.09111

[8] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[9] G. Jianya, S. Haigang, M. Guorui, and Z. Qiming, "A review of multi-temporal remote sensing data change detection algorithms," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 37, no. B7, pp. 757–762, 2008.

[10] A. C. Mondini, F. Guzzetti, P. Reichenbach, M. Rossi, M. Cardinali, and F. Ardizzone, "Semi-automatic recognition and mapping of rainfall induced shallow landslides using optical satellite images," *Remote Sens. Environ.*, vol. 115, no. 7, pp. 1743–1757, Jul. 2011.

[11] M. K. Ridd and J. Liu, "A comparison of four algorithms for change detection in an urban environment," *Remote Sens. Environ.*, vol. 63, no. 2, pp. 95–100, Feb. 1998.

[12] E. F. Lambin and A. H. Strahlers, "Change-vector analysis in multitemporal space: A tool to detect and categorize land-cover change processes using high temporal-resolution satellite data," *Remote Sens. Environ.*, vol. 48, no. 2, pp. 231–244, May 1994.

[13] P. Du, X. Wang, D. Chen, S. Liu, C. Lin, and Y. Meng, "An improved change detection approach using tri-temporal logic-verified change vector analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 278–293, Mar. 2020.

[14] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[15] M. H. Kesikoğlu, Ü. H. Atasever, and C. Özkan, "Unsupervised change detection in satellite images using fuzzy c-means clustering and principal component analysis," *ISPRS-Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vols. 7, pp. 129–132, Oct. 2013.

[16] S. Marchesi and L. Bruzzone, "ICA and kernel ICA for change detection in multispectral remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 2, Jul. 2009, p. 890.

[17] C. Zhang, S. Wei, S. Ji, and M. Lu, "Detecting large-scale urban land cover changes from very high resolution remote sensing images using CNN-based classification," *ISPRS Int. J. Geo-Information*, vol. 8, no. 4, p. 189, Apr. 2019.

[18] C. Wu, B. Du, X. Cui, and L. Zhang, "A post-classification change detection method based on iterative slow feature analysis and Bayesian soft fusion," *Remote Sens. Environ.*, vol. 199, pp. 241–255, Sep. 2017.

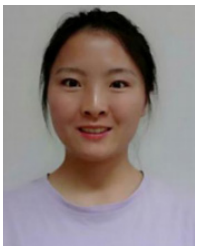
[19] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.

[20] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[21] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi-and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.

[22] A. Lefebvre, T. Corpetti, and L. Hubert-Moy, "Object-oriented approach and texture analysis for change detection in very high resolution images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, vol. 4, Jul. 2008, p. 663.
[23] L. Huang, G. Zhang, and Y. Li, "An object-based change detection approach by integrating intensity and texture differences," in *Proc. 2nd Int. Asia Conf. Informat. Control, Autom. Robot. (CAR)*, vol. 3, Mar. 2010, pp. 258–261.
[24] L. D. Robertson and D. J. King, "Comparison of pixel-and object-based classification in land cover change mapping," *Int. J. Remote Sens.*, vol. 32, no. 6, pp. 1505–1529, 2011.
[25] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.
[26] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.
[27] M. Kampffmeyer, A.-B. Salberg, and R. Jenssen, "Semantic segmentation of small objects and modeling of uncertainty in urban remote sensing images using deep convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2016, pp. 1–9.
[28] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.
[29] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[30] Z. Deng, H. Sun, S. Zhou, J. Zhao, L. Lei, and H. Zou, "Multi-scale object detection in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 3–22, Nov. 2018.
[31] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.
[32] H. Zhang, M. Gong, P. Zhang, L. Su, and J. Shi, "Feature-level change detection using deep representation and feature change analysis for multispectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 11, pp. 1666–1670, Nov. 2016.
[33] Q. Zhao, J. Ma, M. Gong, H. Li, and T. Zhan, "Three-class change detection in synthetic aperture radar images based on deep belief network," *J. Comput. Theor. Nanosci.*, vol. 13, no. 6, pp. 3757–3762, Jun. 2016.
[34] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.
[35] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral-spatial-temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.
[36] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.
[37] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.
[38] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
[39] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[40] H. Li, P. Xiong, J. An, and L. Wang, "Pyramid attention network for semantic segmentation," 2018, *arXiv:1805.10180*. [Online]. Available: http://arxiv.org/abs/1805.10180
[41] J. Chen *et al.*, "DASNet: Dual attentive fully convolutional siamese networks for change detection of high resolution satellite images," 2020, *arXiv:2003.03608*. [Online]. Available: http://arxiv.org/abs/2003.03608
[42] R. Liu, Z. Cheng, L. Zhang, and J. Li, "Remote sensing image change detection based on information transmission and attention mechanism," *IEEE Access*, vol. 7, pp. 156349–156359, 2019.
[43] M. Papadomanolaki, S. Verma, M. Vakalopoulou, S. Gupta, and K. Karantzalos, "Detecting urban changes with recurrent neural networks from multitemporal Sentinel-2 data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.* , Jul. 2019, pp. 214–217.
[44] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.
[45] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, Oct. 2018.
[46] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Springer, 2015, pp. 234–241.
[47] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "Unet++: A nested u-net architecture for medical image segmentation," in *Deep Learn. Med. Image Anal. Multimodal Learn. for Clin. Decis. Support: Springer*, vol. 2018, pp. 3–11.
[48] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *ISPRS-Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, pp. 565–571, May 2018.
[49] C. Benedek and T. Sziranyi, "Change detection in optical aerial images by a multilayer conditional mixed Markov model," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 10, pp. 3416–3430, Oct. 2009.
[50] R. C. Daudt, B. Le Saux, A. Boulch, and Y. Gousseau, "Urban change detection for multispectral Earth observation using convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 2115–2118.
[51] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.



Xueli Peng received the bachelor's degree in geographical information science from Anhui Jianzhu University, Hefei, China, in 2018. She is pursuing the master's degree in cartography and geographic information systems with the Beijing Advanced Innovation Center for Imaging Technology, Capital Normal University, Beijing, China.

Her research interests include deep learning and remote sensing images change detection.



Ruofei Zhong received the Ph.D. degree in geoinformatics from the Chinese Academy of Science, Institute of Remote Sensing Applications, Beijing, China, in 2005.

He is a Professor with Beijing Advanced Innovation Center for Imaging Technology, Key Laboratory of 3D Information Acquisition and Application, College of Resource Environment and Tourism, Capital Normal University, Beijing. His research interests include remote sensing satellite design and data preprocessing, multisensor lidar integration and post-processing, deep learning, and computer vision.



Zhen Li received the bachelor's degree in surveying and mapping engineering from the Shandong University of Science and Technology, Qingdao, China, in 2017. He is pursuing the master's degree in cartography and geographical information engineering with the Beijing Advanced Innovation Center for Imaging Technology, Capital Normal University, Beijing, China.

His research interests include deep learning and remote sensing images semantic segmentation.



Qingyang Li received the bachelor's degree in surveying and mapping engineering from Shandong Jianzhu University, Shandong, China, in 2017, and the master's degree in cartography and geographical information engineering from Capital Normal University, Beijing, China, in 2020, where she is pursuing the Ph.D. degree.

Her research interests include remote sensing data processing and algorithm development.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 11:30:54 UTC from IEEE Xplore. Restrictions apply.